ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 2 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (practical, 100 marks): a Scratch speed-camera app with an IPO chart, a Python turtle-graphics pattern, and a Python data-bundle shop with a function, selection, a loop and a global total.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Scratch for QUESTION 1 and a Python IDE (Python 3) for QUESTIONS 2 and 3. If you do not have a computer, write the programs neatly in your answer book.
3. Save each program with the question number and your name or examination number. Add your name or examination number as a comment on the first line of each Python program.
4. Save your work regularly. Programs are not marked strictly on syntax.

QUESTION 1

A traffic department wants a simple Scratch app to work out speeding fines. The sprite asks for a vehicle's speed and the speed limit of the road. If the vehicle is not speeding, the sprite says "No fine". If it is up to 20 km/h over the limit, the fine is R500. If it is more than 20 km/h over the limit, the fine is R1 500. The sprite then says how far over the limit the vehicle was and the fine.

Use the IPO chart below to help you complete the application:

INPUT	PROCESS	OUTPUT
speed limit	Declare all variables and set them to 0 Ask for the speed; ask for the speed limit over = speed − limit If over ≤ 0: fine = 0, say "No fine" Else if over ≤ 20: fine = 500 Else: fine = 1500 Say over and fine	Message fine

Follow the instructions below to create the Scratch application:

1.1 Open Scratch, create a new project and name it Question1_YourName. (1)

1.2 Choose a sprite to act as a traffic officer. (1)

1.3 Choose a backdrop that shows a road. (1)

1.4 Declare all the variables and give them their starting values. (8)

1.5 Ask for the speed of the vehicle and store the answer. (3)

1.6 Ask for the speed limit and store the answer. (3)

1.7 Calculate how far over the limit the vehicle was. (2)

1.8 Use if … then … else blocks to decide the fine. (10)

1.9 Display the result. (3)

1.10 Test the program with these values and write down what the sprite says: 58 km/h in a 60 zone; 75 km/h in a 60 zone; 145 km/h in a 120 zone. (3)

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QUESTION 2

Write a Python program that uses the turtle graphics library to draw a pattern of 36 hexagons, each one turned a little from the last, so that together they form a round flower-like pattern. Each hexagon has sides of 100 pixels.

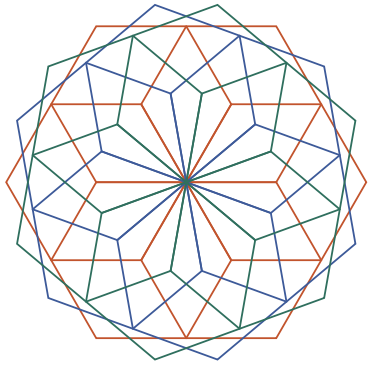


FIGURE 2.1: Example of the hexagon pattern (your colours may differ)

2.1 Create a new file called Question2_YourName.py and add your name or examination number as a comment on the first line. (1)

2.2 Import the turtle (and random) modules and create a turtle object named pen. (2)

2.3 Set the drawing speed to 0 (fastest). (2)

2.4 Set the thickness of the lines to 3 pixels. (2)

2.5 Use a loop that draws the hexagon 36 times. (5)

2.6 Use a second (inner) loop to draw each hexagon: 6 sides of 100 pixels, turning 60° after each side. (5)

2.7 Turn the turtle 10° after each hexagon, and give each hexagon a random colour. (5)

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QUESTION 3

A cellphone shop sells data bundles. The price per GB depends on the size of the bundle: up to 5 GB costs R30 per GB; from 6 GB to 20 GB costs R25 per GB; more than 20 GB costs R20 per GB. A customer who buys 3 or more bundles at once gets a 10% discount. The shop wants a Python program that works out each customer's cost, lets the cashier serve as many customers as needed, and shows the total sales at the end.

The following steps show how the program should be used:

```
Enter customer name: Thandi
Enter bundle size (GB): 10
Enter number of bundles: 2
Receipt for Thandi: 2 x 10 GB at R25 per GB = R500.00
Serve another customer? (yes/no): yes
Enter customer name: Sipho
Enter bundle size (GB): 3
Enter number of bundles: 4
Receipt for Sipho: 4 x 3 GB at R30 per GB = R324.00 (10% discount)
```

```
Serve another customer? (yes/no): no
Total sales: R824.00
```

Follow the instructions below to create the application:

3.1	Create a new Python file called Question3_YourName.py.	(1)
3.2	Declare a global variable named total_sales and set it to 0.	(2)
3.3	Create a function named sell_bundle() with no parameters. Inside it, ask for the customer's name, the bundle size in GB and the number of bundles.	(6)
3.4	Use if, elif and else to choose the price per GB from the bundle size.	(8)
3.5	Give a 10% discount when 3 or more bundles are bought.	(4)
3.6	Calculate the customer's cost and display the receipt.	(4)
3.7	Use a loop that lets the cashier serve more customers until the answer is "no".	(7)
3.8	Add each customer's cost to total_sales.	(4)
3.9	Call the function inside the loop and display the total sales at the end.	(4)
3.10	Test the program with the data shown in the example above.	(2)
3.11	Save the program and print (or neatly present) the code.	(1)

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TOTAL: 100

END OF QUESTION PAPER